

WORMWOOD (ARTEMISIA ABSINTHIUM): A BITTER HERB APPLIED IN PHYTOTHERAPY

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RESUME

Bitters are a heterogeneous class of plant-derived compounds defined by their ability to activate bitter taste receptors (T2Rs) located mainly on the posterior tongue. Stimulation of these receptors initiates a neurophysiological reflex via the glossopharyngeal and vagus nerves, enhancing gastric, biliary, and pancreatic secretions. Medicinal bitters thus act not merely through taste perception but by eliciting a reflex arc that optimizes digestive function, appetite regulation, and gastrointestinal motility.

Pharmacologically, bitters are beneficial in hypochlorhydria, anorexia, and dyspeptic conditions, though excessive intake may cause mucosal irritation and inhibition of gastric secretion. Beyond digestion, they may exert mild systemic effects through parasympathetic activation. Due to their capacity to increase gastric acidity, bitters are contraindicated in gastritis, peptic ulcer disease, and duodenal ulceration.

Botanically, a bitter **wormwood (Artemisia absinthium L.)** is an aromatic perennial native to Eurasia and naturalized worldwide. The aerial parts contain sesquiterpene lactones (e.g., absinthin, artabsin) and 0.3–0.5% volatile oil, primarily composed of the neuroactive monoterpene thujone in two isomeric forms, (–)-thujone and (+)-isothujone. At therapeutic doses (≈1 g infusion or tincture), wormwood functions as an aromatic bitter, stimulating gastric and biliary secretion. At higher doses, thujone produces neurotoxic effects, including seizures, hallucinations, and delirium.

Historically, wormwood and related *Artemisia* species were valued for their digestive, choleric, and antiparasitic actions, and were incorporated into liqueurs such as absinthe and vermouth. Due to thujone toxicity, pure wormwood liquors are now prohibited. *A. pontica* and *A. maritima* are preferred for vermouth production due to their lower thujone content.

Pharmacopoeial standards require a minimum volatile oil content of 0.3%. The recommended daily dose is 2–3 g of the dried herb, administered for short-term use to enhance appetite and digestion.

Key Words: Bitters, Wormwood, *Artemisia absinthium*.

აბზინდა (*Artemisia absinthium*) და მისი გამოყენება ფიტოთერაპიაში

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მწარეები წარმოადგენს ფიტოქიმიური ნივთიერებების მრავალფეროვან ჯგუფს, რომლებიც ახდენენ ენის უკანა ზედაპირზე მდებარე მწარე გემოს რეცეპტორების (T2R) აქტივაციას. მათი სტიმულაცია იწვევს ნერვულ რეფლექსს ყლაპვისა და ვაგუსის ნერვების მეშვეობით, რაც აძლიერებს კუჭის, ნაღვლისა და პანკრეასის სეკრეციას. ამიტომ მწარეები მოქმედებენ არა მხოლოდ გემოს აღქმის გზით, არამედ რეფლექსური მექანიზმით, რომელიც აუმჯობესებს საჭმლის მონელებასა და მადას.

ფარმაკოლოგიურად, მწარეები გამოიყენება ჰიპოქლორჰიდიის, ანორექსიისა და დისპეფიის დროს, თუმცა ზედმეტი დოზა იწვევს ლორწოვანი გარსის გაღიზიანებას და კუჭის სეკრეციის დაქვეითებას. გარდა ამისა, მათ შეიძლება იმოქმედონ პარასიმპათიკური სისტემის მეშვეობით, რამაც შესაძლოა გამოიწვიოს გულისცემის დროებითი შემცირება. მწარეები უკუნაჩვენებია გასტრიტის, წყლულოვანი დაავადებისა და დუოდენალური წყლულის დროს.

ბოტანიკურად, **აბზინდა (*Artemisia absinthium* L.)** არის არომატული მრავალწლოვანი მცენარე, რომელიც ფართოდ არის გავრცელებული ევრაზიასა და სხვა რეგიონებში. მისი მიწისზედა ნაწილები შეიცავს სესკვიტერპენულ ლაქტონებს (მაგ. აბსინთინი, არტაბსინი) და დაახლოებით 0.3–0.5% ეთერზეთს, რომლის 70%-მდე შეადგენს თუჯონის ორი სტერეოიზომერი — (–)-თუჯონი და (+)-იზო-თუჯონი. მცირე დოზით (დაახლოებით 1 გ/ინფუზია ან ნაყენი) აბსინთი მოქმედებს როგორც არომატული მწარე, აუმჯობესებს კუჭისა და ნაღვლის სეკრეციას. დიდი დოზები იწვევს ნეიროტოქსიკურ ეფექტებს — თავის ტკივილს, კრუნჩხვებს, ჰალუცინაციებსა და დელირიუმს.

ისტორიულად, აბზინდა და *Artemisia*-ს სხვა სახეობები გამოიყენებოდა საჭმლის მონელების, ნაღვლის გამოდენისა და პარაზიტული ინფექციების

სამკურნალოდ, აგრეთვე ლიქიორების (აბსენტის, ვერმუტი) დასამზადებლად. თუჯონის ტოქსიკურობის გამო მისი სუფთა ალკოჰოლური ექსტრაქტები აკრძალულია.

ფარმაკოპეის მოთხოვნით, ნედლეული უნდა შეიცავდეს მინიმუმ 0.3% ეთერზეთს. რეკომენდებული დღიური დოზაა 2–3 გ გამშრალი ბალახისა, მოკლევადიანი კურსით მადასა და მონელების გასაუმჯობესებლად.

საკვანძო სიტყვები: მწარეები, აბზინდა *Artemisia absinthium*

Bitters are a diverse group of phytochemical substances characterized by their pronounced ability to activate bitter taste receptors (T2R family) located primarily on the posterior aspect of the tongue. Upon stimulation of these G-protein-coupled receptors, a complex neuro-physiological reflex is initiated via afferent fibers of the glossopharyngeal and vagus nerves, ultimately influencing gastric, biliary, and pancreatic secretory functions.

Medicinal bitters exert their therapeutic effect not solely through gustatory perception but by triggering a reflex arc that enhances digestive function. The initial stimulation of oropharyngeal bitter receptors elicits a parasympathetic response, resulting in increased secretion of saliva, gastric acid, bile, and digestive enzymes. This reflex pathway also promotes improved gastrointestinal motility and appetite regulation.

At pharmacologically optimal concentrations, bitters enhance digestive efficiency and stimulate appetite, making them valuable in conditions of hypochlorhydria or anorexia of non-organic origin. Conversely, excessive intake can lead to mucosal irritation and attenuation of gastric secretion, mediated by direct desensitization or inhibitory feedback mechanisms at the level of the gastric mucosa.

Beyond the gastrointestinal tract, bitters exert mild systemic effects. Reflex activation of the autonomic nervous system may transiently reduce heart rate and cardiac stroke volume through vagal stimulation. Chronic administration, however, can lead to sensory adaptation and aversive responses, sometimes accompanied by anorexia or gastrointestinal discomfort. Therefore, bitters are typically indicated for short-term therapeutic regimens.

The bitterness of herbal preparations cannot be effectively masked by sucrose or other sweetening agents, as such additives do not inhibit receptor activation. Adverse effects are generally mild and self-limiting; however, headaches or nausea may occur in sensitive individuals. Overdosage can induce emesis. Because of

their stimulatory influence on gastric acid secretion, bitters are contraindicated in patients with peptic ulcer disease, gastritis, or duodenal ulceration.

Many cultures recognise the value of bitter substances in promoting digestive function and general health. In Holland, older people would celebrate the bitter hour in the early evening when they would partake of bitter food and drink to support their fading digestive powers. In India, it is said that those with liver problems seek bitter-tasting substances. In Africa, the medicinal value of bitter herbs, particularly as digestive stimulants, is commonly recognised in traditional-medical systems.(1) Bitter drinks taken before meals are still called aperitifs. In the early 20th century, it was still widely accepted in medical and scientific circles that bitters promoted digestion. Even Pavlov was said to have acknowledged this connection (2). However, this was a time when such assumptions were being subjected to scientific scrutiny. In 1915, the American physiologist Carlson and co-workers published a study entitled 'The Supposed Action of the Bitter Tonics on the Secretion of Gastric Juice in Man and Dog'.(2) The group found that bitters either applied to the mouth or directly to the stomach, produced no change in the acidity and pepsin concentration of the gastric juice produced prior to food actually being in contact with the stomach. Despite the fact that this study had a number of methodological flaws, notably that gastric secretions were not tested under the stimulus of actual contact with food, it was largely interpreted as discrediting the concept of bitters as digestive stimulants. However, work published also in 1915 by Moorhead, a colleague of Carlson, demonstrated a radically different activity profile for bitters.(3) The conclusions that could be drawn from this early research are several:

- Bitters markedly increase appetite only if a cachectic, malnourished or debilitated state exists in the body.
- Similarly, bitters increase digestive power mainly when it is below optimum, as in a state of cachexia.
- Experiments with bitters should involve actual feeding; that is, the presence of food in the stomach is important for their activity.
- At normal doses, bitters act in the mouth; that is, tasting optimises their activity.

Bitters may be classified based on their predominant phytochemical constituents and associated pharmacological profiles:

1. **Simple Bitters** – Contain primarily bitter glycosides without significant aromatic or astringent components; e.g., *Gentiana lutea* (gentian), *Menyanthes trifoliata* (bogbean), *Centaureum erythraea* (centaury).

2. **Aromatic Bitters** – Characterized by the presence of volatile oils and essential terpenoids, contributing both to bitterness and aromaticity; e.g., *Angelica archangelica* (angelica root), *Cnicus benedictus* (blessed thistle), *Citrus aurantium* (bitter orange peel), *Artemisia absinthium* (wormwood).

3. **Astringent Bitters** – Contain tannins that confer both bitterness and astringency; e.g., *Cinchona officinalis* (cinchona bark), *Marsdenia condurango* (condurango bark).

4. **Acrid Bitters** – Possess pungent or irritant constituents such as alkyl phenols or gingerols; e.g., *Zingiber officinale* (ginger), *Alpinia galanga* (galangal).

As mentioned above, aromatic bitters are those bitter-tasting herbs that also have a strong aromatic (volatile oil) component. This makes them both stimulating to the digestion and gently warming/spasm-relieving.

Wormwood (*Artemisia absinthium* L.) is a perennial, aromatic shrub with fibrous roots native to arid and semi-arid regions of Eurasia (7), and naturalized in North and South America (8) as well as New Zealand. The **leaves and flowering tops** are collected from both wild-growing and cultivated plants for medicinal and industrial use (6)



Picture 1. Wormwood (*Artemisia absinthium* L.)

Wormwood's relative mugwort was traditionally used as a remedy for a variety of complaints, especially those of a gynaecological nature, and so the wormwood genus bears the name of the Greek goddess of childbirth, Artemis(9).

The stems of ***Artemisia absinthium* L** are straight, growing to 0.8–1.2 m (and rarely over 1.5 metres) tall, grooved, branched, and silvery-green. Leaves are spirally arranged, greenish-grey colored above, white below, covered with silky silvery-white trichomes, and bearing minute oil-producing glands. The basal leaves are up to 250 mm long, bi- to tripinnate with long petioles, with the cauline leaves (those on the stem) smaller, 50–100 mm long, less divided, and with short petioles. The uppermost leaves can be both simple and sessile (without a petiole).

Flowers are pale yellow, tubular, and clustered in spherical bent-down heads (capitula), which are in turn clustered in leafy and branched panicles. Flowering occurs from early summer to early autumn; pollination is anemophilous. The fruit is a small achene. Seed dispersal occurs by gravity.



Picture 2. Leaves and flowers of Wormwood (*Artemisia absinthium* L.)

The plant is characterized by a **penetrating, aromatic odor** and a **strongly bitter, spicy taste**.

The **bitter principles** present in the aerial parts are primarily **sesquiterpene lactones**, which occur as **monomeric compounds** (e.g., *artabsin*) and **dimeric forms** (e.g., *absinthin*). The herb contains approximately **0.3–0.5% volatile oil**, of which up to **70%** consists of **two stereoisomeric forms of thujone** — **(-)-thujone** and **(+)-isothujone**. These monoterpene ketones are responsible for the herb's characteristic **fresh, camphoraceous, and spicy aroma**.

Thujone is neurotoxic and its presence in liqueurs such as absinthe apparently caused widespread toxicity and abuse syndromes in the early 20th century(10). The first sign of toxicity is a headache.

Wormwood is traditionally used to stimulate appetite and to treat dyspeptic complaints, including gastritis and gall bladder ailments (biliary dyskinesia). The long-standing use of aqueous and ethanolic preparations of *Absinthii herba*, pharmacological studies and current findings of physiological properties justify the use of *Artemisia absinthium* for the treatment of loss of appetite and for the symptomatic treatment of dyspepsia and mild spasmodic disorders of the gastrointestinal tract. The plant is also described as a choleric (13).

Extraction with 70% (v/v) ethanol 3 times daily, each single dose equivalent to 1 g herbal substance to improve

appetite and stimulate digestion in the case of hypoacidity and chronic gastritis was found to be effective (14). A typical traditional dose of wormwood practiced in many villages of the Kashmir valley is 3 cups daily of a tea made by steeping 10 grams of wormwood in 250 ml of hot water. Most reputable sources recommend a range from 1-1.5g (15), (16) (5),(17).of dried herb as an infusion up to three times per day (maximum of 3g/day), for no longer than 3-4 weeks after which time an aversion to the tea/preparations is likely to naturally occur (18) Wormwood pills 0.1-0.2g/dose (18), however, caution is required to avoid over-dosing (16). For Crohn's disease it is recommended to 1.5g/day (500mg pill three times a day) (19). After intake of a concentrated infusion of *Absinthii herba*, a male developed dizziness, atony, and tremor of the legs (20) and it is stated that excessive doses of *Absinthii herba* preparations may cause vomiting, severe diarrhea, retention of urine or dazed feelings (21). Overdose of alcoholic *Absinthii herba* preparations or the use of the essential oil may cause CNS disturbances which can lead to convulsions and ultimately to unconsciousness and death (21), (22). Cases with severe intoxications in humans have been reported after consumption of essential oil rich in

thujone (23). Excessive dosing, long-term use or use of the essential oil could lead to toxicity (17).

Side effects may include brain damage, convulsions, even death; insomnia, intestinal cramps, nausea, restlessness,

seizures, stomach cramps, tremors, urine retention, vertigo and vomiting (24).

Traditionally, preparations derived from the aerial parts of various *Artemisia* species have been used not only as **aromatic bitters** to stimulate appetite and digestive secretions, but also in the management of **infectious diseases** such as **malaria** and **hepatitis**.

Artemisia absinthium L. is used to treat epilepsy, gastric problems, enlargement of spleen, urinary disorders and for

wound healing (11). It is topically used to treat skin disorders. It has long been used to induce labor, and more recently to treat epilepsy and spasms. Mahmoudi (12) suggested the antidepressant activity of methanolic extract from aerial parts of *A. absinthium* at flowering stage.

Throughout history wormwood was particularly favoured for its fever-reducing properties as well as having a range of antipathogenic and anti-inflammatory properties. However, it is inherently very powerful and should be applied with more caution and under closer supervision

At **low doses** (e.g., approximately **1 g** of the crude drug administered as an infusion or tincture, wormwood exhibits **aromatic bitter activity**, enhancing gastric and biliary secretion. However, with increasing dosage, the **neurotoxic effects of thujone** become evident, manifesting as **excess salivation, hyperemia of the mucous membranes, and congestion of the pelvic viscera**.

The **production of pure wormwood-based liquors** (e.g., absinthe) is **prohibited in many countries** due to the risk of **thujone-induced neurotoxicity** and **addictive potential**. Thujone has been shown to **potentiate and modify the effects of ethanol**, and **chronic intoxication** may result in **cerebral dysfunction**, characterized by **epileptiform seizures, delirium, and hallucinations** — a condition historically termed *absinthism*.

Roman wormwood (*Artemisia pontica* L.) is preferred for the manufacture of **vermouth wines** because of its **milder aroma** and **lower thujone content**, which enhances flavor without excessive bitterness. The **leaves of maritime wormwood (*Artemisia maritima* L.)** are also occasionally employed for similar purposes.

According to pharmacopoeial standards, the **crude drug** should contain **not less than 0.3% volatile oil** and exhibit a **relative bitterness value of at least**

15,000. The **recommended average daily dose** is **2–3 g** of the dried aerial parts (6).

In conclusion, bitters—particularly aromatic species such as *A. absinthium*—demonstrate a clear physiological capacity to stimulate digestive processes and modulate autonomic activity. However, their therapeutic efficacy depends on dosage precision, proper preparation, and awareness of toxicity thresholds.

In conclusion, Bitters represent a pharmacologically active class of phytochemicals that elicit digestive stimulation primarily through activation of gustatory T2R receptors and subsequent autonomic reflexes mediated by the glossopharyngeal and vagus nerves. Among these, *Artemisia absinthium* (wormwood) exemplifies the aromatic bitters, exerting choleric, stomachic, and spasmolytic effects attributable mainly to sesquiterpene lactones and volatile monoterpenes such as thujone. At therapeutic doses, wormwood enhances gastric and biliary secretions and improves appetite in cases of hypochlorhydria or dyspeptic disorders. However, excessive intake or prolonged use poses significant neurotoxic risk due to thujone-induced central nervous system excitation. Thus, while bitters retain validated physiological activity as digestive stimulants, their safe and effective application requires precise dosing, short-term administration, and awareness of contraindications such as peptic ulceration or gastritis.

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